ChemCorp Industries S.A.
MEETING MINUTES

R&D Meeting — Bio-based solvent formulation development

Ref: CRI-RD-2024-003

Date	2024-01-18	**Time**	09:00–12:30
Location	Innovation Room, Villeurbanne site	**Ref.**	CRI-RD-2024-003
Chair	Dr. Martine Chabrol, R&D Director	**Secretary**	Mr Antoine Ferretti, Project Manager

ATTENDEES

Name	Role / Department	Attendance
Dr. Martine Chabrol	R&D Management	Present
Mr Antoine Ferretti	R&D — Formulation	Present
Ms Sonia Blanc	R&D — Synthesis	Present
Mr Kévin Lecomte	R&D — Analytical	Present
Ms Clara Nguyen	Product Marketing	Present
Mr Bertrand Vidal	Production / Processes	Present
Dr. Paul Renaud	Regulatory / REACH	Apologies — represented by Mr Ferretti

AGENDA

1. Progress update on the BioSolv-3 project (bio-based industrial solvent formulation)
2. Pilot test results on reactor unit R-103
3. Competitive analysis — new entrants in the green solvents market
4. Validation schedule and Q2 2024 patent filing
5. Any other business

PROCEEDINGS

1. BioSolv-3 progress update

Dr. Chabrol opened the meeting at 09:05 and restated the objectives of the BioSolv-3 project: to develop a range of industrial solvents derived more than 70% from renewable raw materials, with performance equivalent to conventional solvents for degreasing and paint formulation applications.

Mr Ferretti presented the current status: formulation F-12C (a blend of isosorbide acetate / methyl lactate / limonene) reaches a KB solvency value of 42, below the target of 50. Two optimisation routes were identified: (a) increasing the limonene fraction from 15% to 22%, and (b) partially substituting the methyl lactate with ethyl lactate to improve miscibility.

Ms Blanc reported thermal stability problems: formulation F-12C shows a tendency to develop colour (APHA above 50) after 48h at 60°C, probably linked to oxidation of the limonene. She proposed evaluating the addition of a phenolic antioxidant (BHT at 200 ppm) and working under an inert atmosphere during packaging.

Mr Lecomte (analytical) presented the GC-MS characterisation data: the purity of the pilot batch is compliant, but traces of residual lactic acid (180 ppm) were detected, requiring an additional purification step. Proposal: passing the material over a basic ion-exchange resin.

2. Pilot test results — Reactor R-103

Mr Vidal (Production) presented the results of the pilot campaign of 8-12 Jan 2024 on reactor R-103 (50 L). Reaction yield: 87.3% (target: above 90%). The pressure observed at the end of the reaction (4.2 bar) slightly exceeds the nominal value of 4.0 bar — a check of safety valve SV-103B is scheduled for 25 Jan.

The 85°C temperature setpoint is difficult to maintain: sensors TC-103A and TC-103B show a deviation of ±3°C at steady state. Mr Vidal recommended recalibrating the sensors and evaluating an adaptive PID controller.

Mass balance: 3.2 kg of unidentified by-products collected in the sump. Mr Lecomte will analyse these fractions by HPLC during the week of 22 Jan. Results are expected for the next weekly update.

Ms Blanc proposed testing heterogeneous acid catalysis (H-ZSM-5 zeolite) to replace the current homogeneous catalyst, to make recovery easier and improve selectivity. To be confirmed with the supplier Zeochem before trials begin.

3. Competitive analysis — Green solvents

Ms Nguyen (Marketing) presented market intelligence gathered over Q4 2023: at least 4 competitors launched or announced bio-based solvent ranges between September and December 2023. Key points: Greensolv AG (Switzerland) filed patent EP3812450 covering a process for purifying methyl lactate by membrane distillation — Mr Ferretti confirmed that our ion-exchange resin approach is different and not covered.

Observed pricing: premium bio-based solvents at 2.5–3.5× the price of petrochemical equivalents among competitors. ChemCorp is targeting a position at 1.8–2.2× thanks to the optimised process.

Discussion: Ms Nguyen proposed prioritising regulated markets (cosmetics, pharmaceuticals, food-contact cleaning) where customers accept higher prices in exchange for documented bio-based content. Mr Vidal noted the capacity constraints: 200 t/year maximum on the pilot unit, which limits how far heavy industrial markets can be addressed.

4. Validation schedule and patent filing

Schedule approved in the meeting: (1) Finalise the optimised F-12C formulation — 15 March 2024. (2) 200 L scale-up pilot campaign — April 2024. (3) Validation dossier (stability, compatibility, performance) — May-June 2024. (4) PCT patent filing — 30 June 2024 (firm Lebret & Associés). (5) Industrialisation and first commercial launch — Q1 2025.

Mr Ferretti was appointed project coordinator (project manager). A monthly review point is scheduled for the 3rd Thursday of each month, 09:00-10:00, Innovation Room.

2024 R&D budget for BioSolv-3: €380k allocated, of which €120k for analytical work and external testing, €85k for patent costs, and €175k for pilot and raw material costs.

Minutes CRI-RD-2024-003 — Action plan

ACTION PLAN

Owner	Action to be taken	Due date
Ms Blanc	Prepare the BHT antioxidant trial plan + packaging under N■	2024-02-01
Mr Lecomte	Analyse the R-103 reactor by-product fractions by HPLC	2024-01-26
Mr Ferretti	Contact the Lebret firm to scope the PCT patent filing	2024-01-25
Mr Vidal	Schedule recalibration of sensors TC-103A/B and the SV-103B check	2024-01-25
Ms Nguyen	Finalise the market study and price positioning proposal	2024-02-15
Dr. Chabrol	Approve the 2024 BioSolv-3 budget with the Finance Director	2024-01-30

Next meeting: 15 February 2024, 09:00, Innovation Room

These minutes will be deemed approved unless written comments are received within 5 working days of circulation.